



CV项目

图像分割

课程内容

🕒 图像分割概述

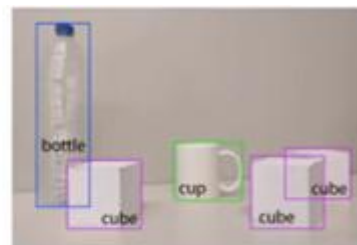
🕒 FCN、 **UNet**

🕒 DeepLab v1、 v2、 v3

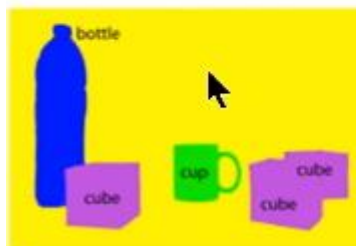
图像分割概述



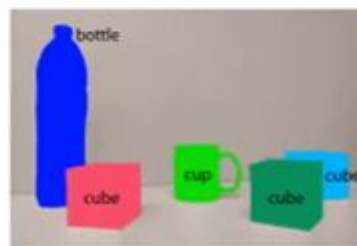
(a) Image classification



(b) Object localization



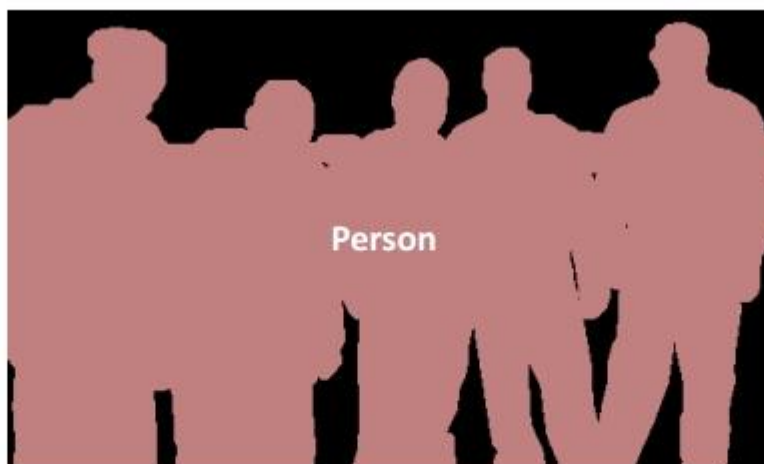
(c) Semantic segmentation



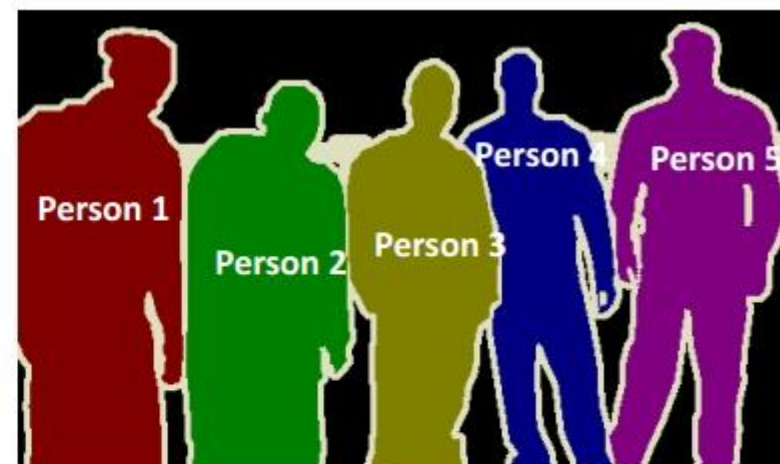
(d) Instance segmentation



Object Detection



Semantic Segmentation



Instance Segmentation



图像分割概述

⌚ 普通分割

⌚ 将分属不同物体的像素区域分开；比如前景和背景的分割开。

⌚ 语义分割

⌚ 在普通分割的基础上，分类出每个区域的语义，也就是给出这个区域具体是什么物体。

⌚ 实例分割

⌚ 在语义分割的基础上，给每个物体以一个具体的编号。

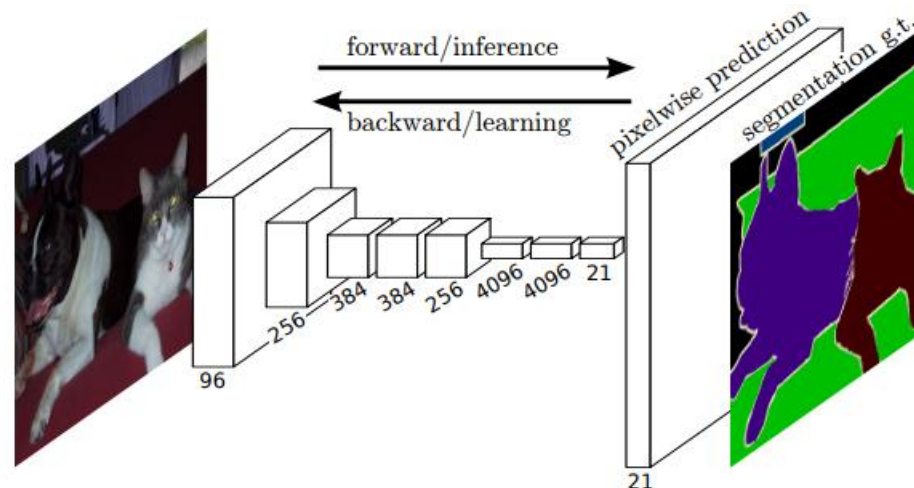
语义分割

⌚ 在CNN图像语义分割中，不管是原始的FCN还是比较新的DeepLab v3+都是按照如下套路进行操作的：

- ⌚ 1. 下采样 + 上采样：Convolution + DeConvolution / Resize
- ⌚ 2. 多尺度特征融合：特征逐点相加 / 特征channel维度拼接
- ⌚ 3. 获取像素级别的segment map：对每个像素点进行类别判断
- ⌚ 4. 其实结构类似：自编码神经网络(AE)

⌚ Fully Convolutional Networks(FCN)

- ⌚ **全卷积网络**，将所有的全连接层替换为卷积层，支持任意图像大小的输入，并且不会产生由于全连接网络丢失原图像的空间位置特征信息；
- ⌚ **反卷积层(DeConvolution)**，也叫做转置卷积(Transposed Convolution)，其作用是将卷积提取出来的特征图上采样映射回原图大小来做分割。
- ⌚ 提出**跳跃式链接(Skip-Layer)**改善上采样粗糙的情况；
- ⌚ <https://arxiv.org/pdf/1411.4038.pdf>



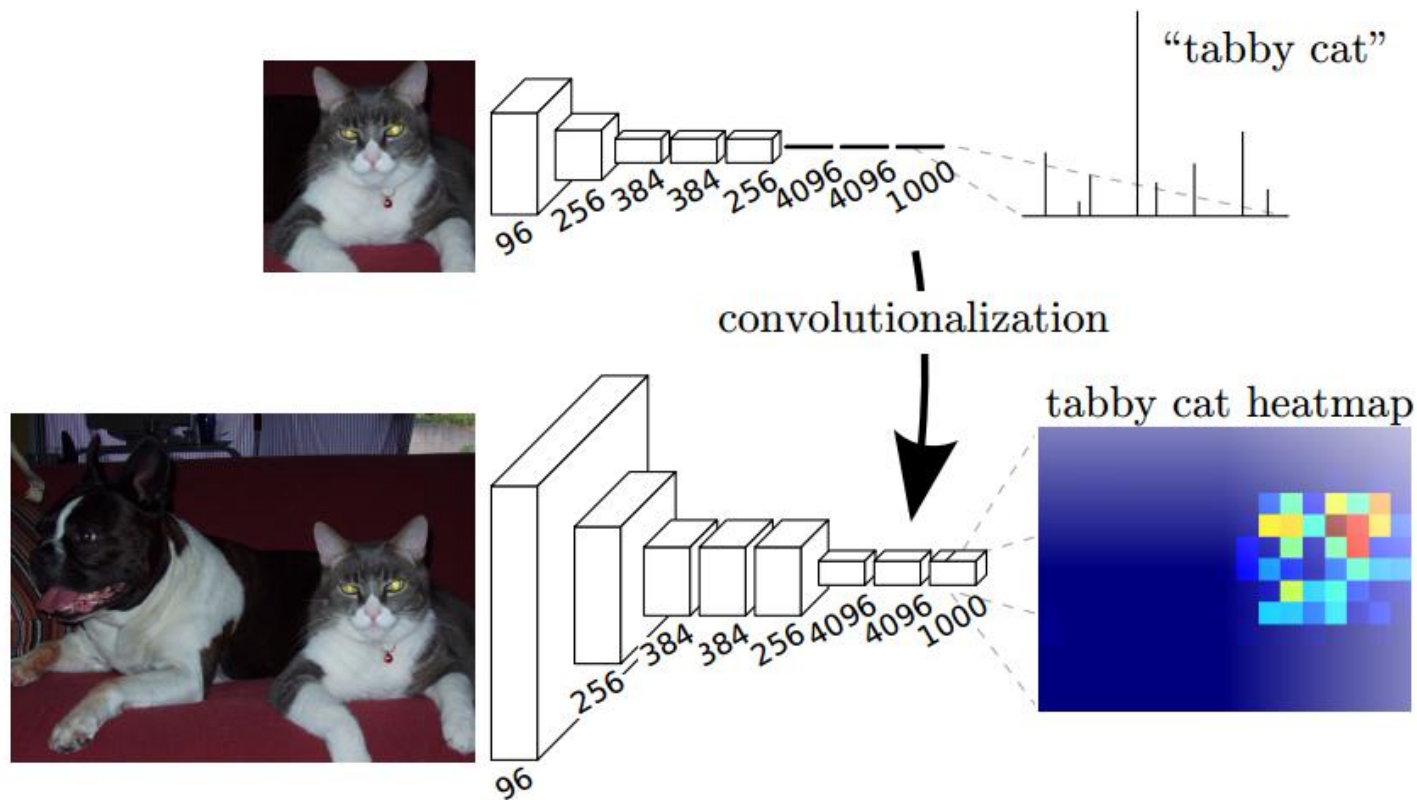


Figure 2. Transforming fully connected layers into convolution layers enables a classification net to output a heatmap. Adding layers and a spatial loss (as in Figure 1) produces an efficient machine for end-to-end dense learning.

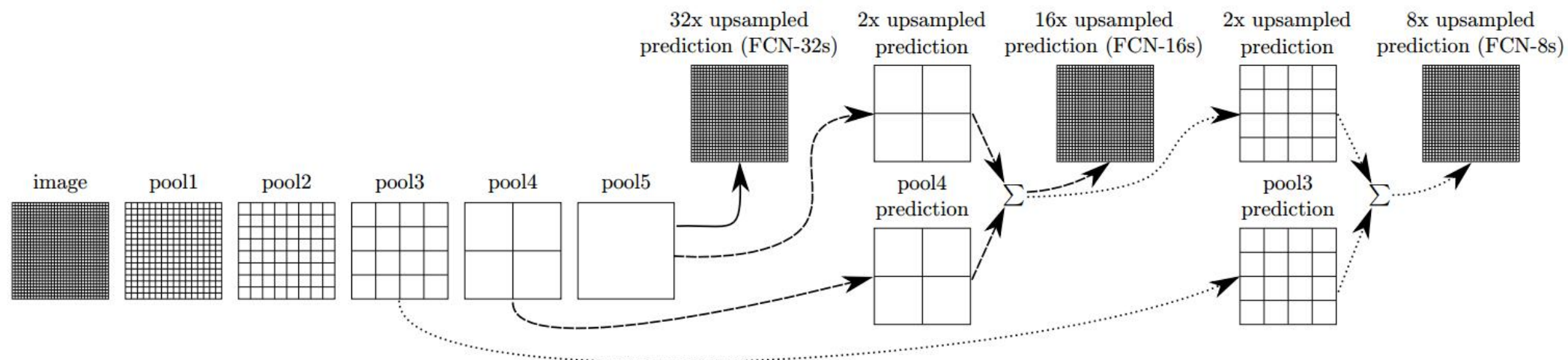
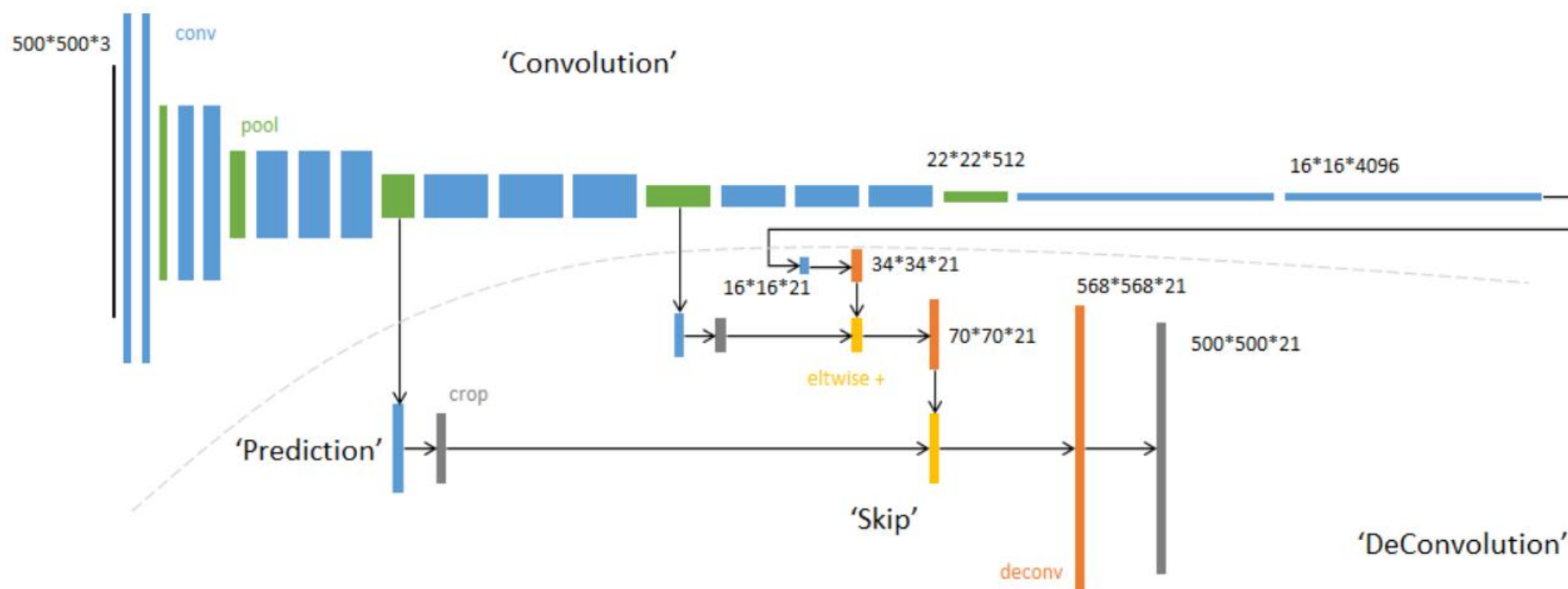


Figure 3. Our DAG nets learn to combine coarse, high layer information with fine, low layer information. Layers are shown as grids that reveal relative spatial coarseness. Only pooling and prediction layers are shown; intermediate convolution layers (including our converted fully connected layers) are omitted. Solid line (FCN-32s): Our single-stream net, described in Section 4.1, upsamples stride 32 predictions back to pixels in a single step. Dashed line (FCN-16s): Combining predictions from both the final layer and the `pool4` layer, at stride 16, lets our net predict finer details, while retaining high-level semantic information. Dotted line (FCN-8s): Additional predictions from `pool3`, at stride 8, provide further precision.

FCN构架图例

蓝色	绿色	黄色	橙色	灰色
卷积层	池化层	逐像素相加	反卷积层	剪裁层
	获取特征	融合结果	上采样	统一尺寸



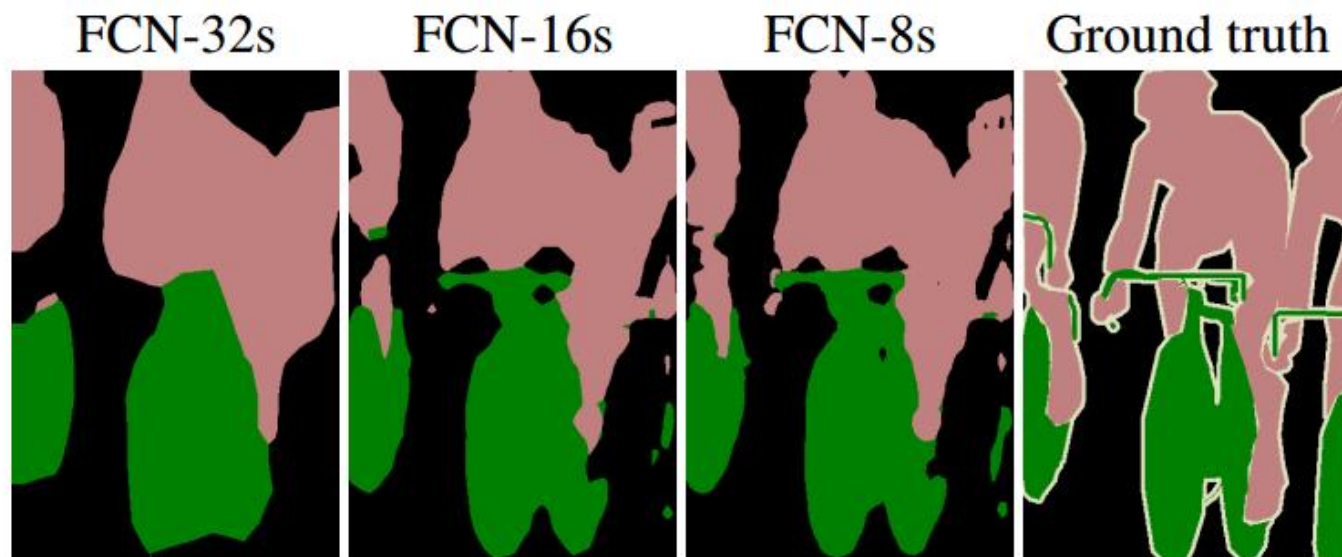


Figure 4. Refining fully convolutional nets by fusing information from layers with different strides improves segmentation detail. The first three images show the output from our 32, 16, and 8 pixel stride nets (see Figure 3).

	pixel acc.	mean acc.	mean IU	f.w. IU
FCN-32s-fixed	83.0	59.7	45.4	72.0
FCN-32s	89.1	73.3	59.4	81.4
FCN-16s	90.0	75.7	62.4	83.0
FCN-8s	90.3	75.9	62.7	83.2

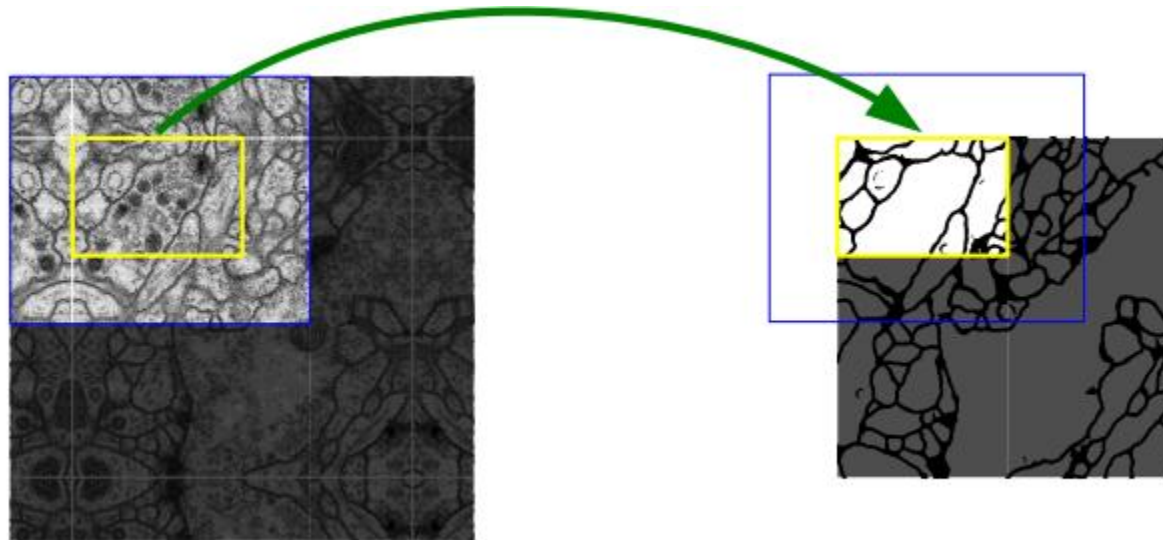
UNet

⌚ UNet属于FCN的一种变体，初期主要应用于生物医学图像方面，后来广泛应用于语义分割的各个方面，包括：卫星图像分割、工业瑕疵检测等。

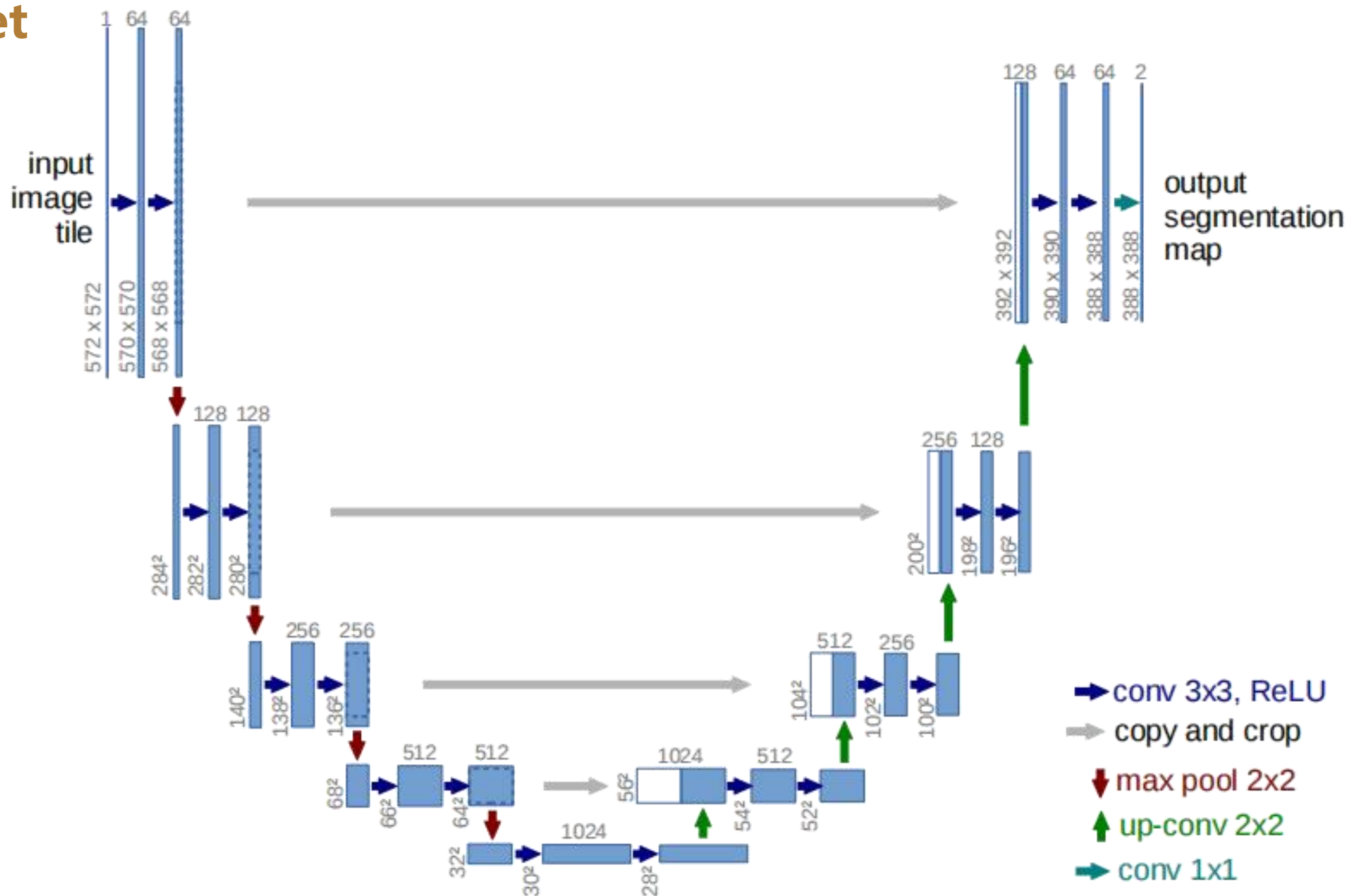
⌚ 论文地址：<https://arxiv.org/abs/1505.04597>

⌚ 特点：

- ⌚ Encoder-Decoder结构
- ⌚ skip-connection结构
- ⌚ Upsampling上采样结构



UNet



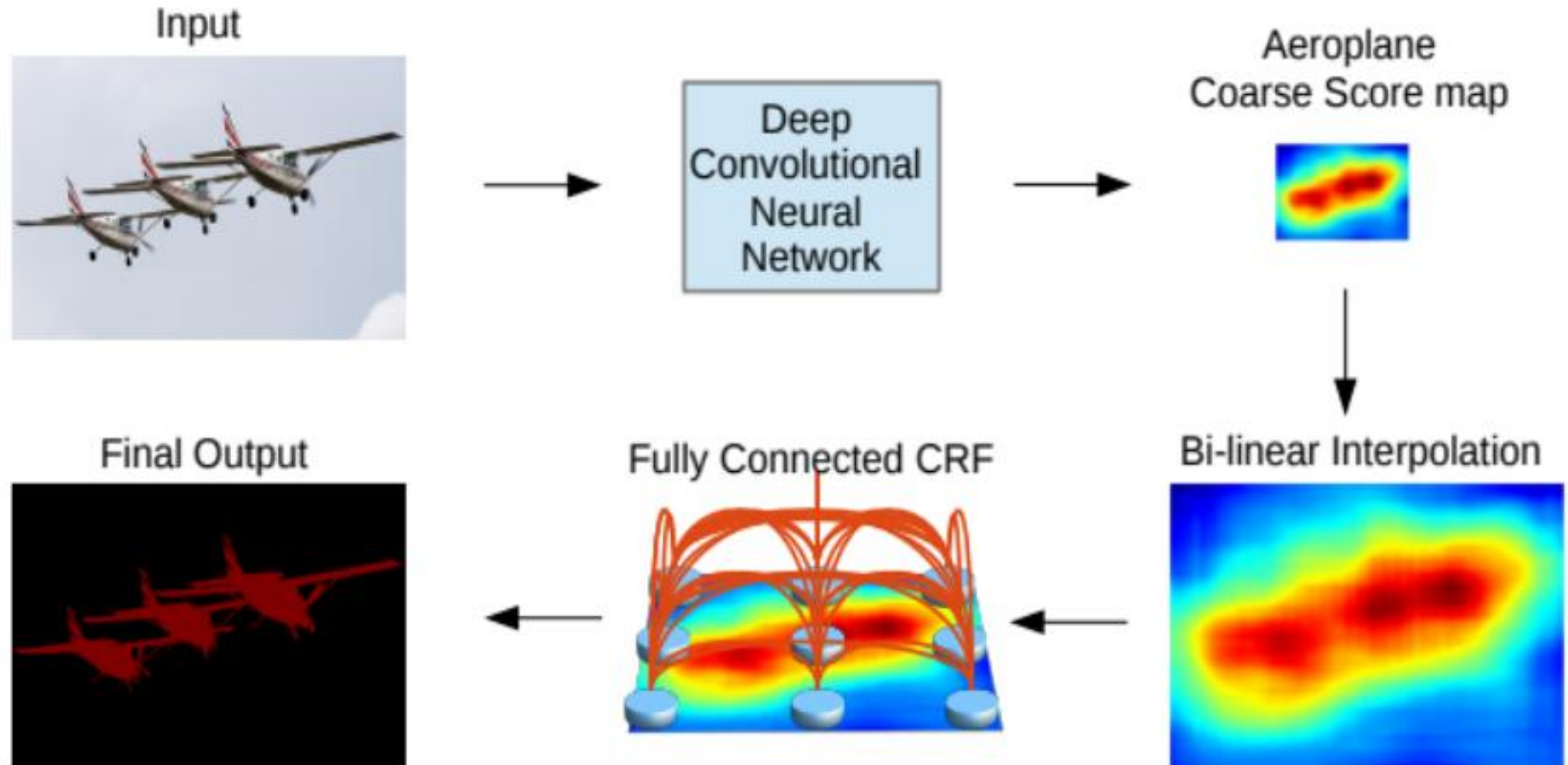
Deep Lab

⌚ Deep Lab v1: Semantic Image Segmentation with Deep Convolutional Nets and Fully Connected CRFs

⌚ <https://arxiv.org/pdf/1412.7062.pdf>

⌚ **Fully Connected CRF**

Deep Lab v1



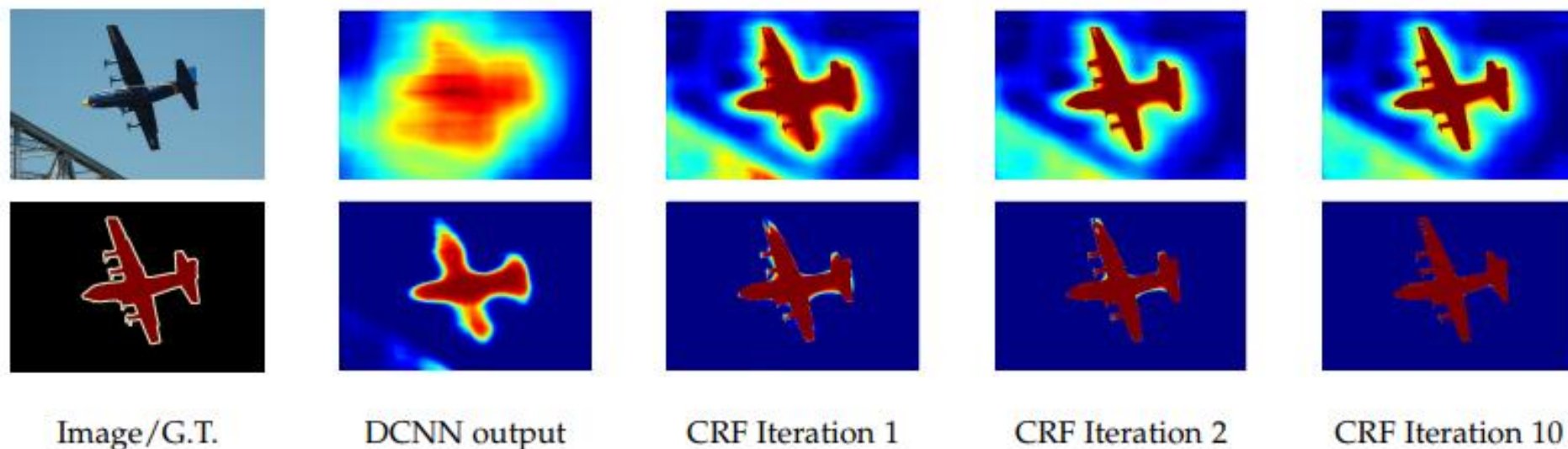


Fig. 5: Score map (input before softmax function) and belief map (output of softmax function) for Aeroplane. We show the score (1st row) and belief (2nd row) maps after each mean field iteration. The output of last DCNN layer is used as input to the mean field inference.

DeepLab v1

Method	bkg	aero	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	mbike	person	plant	sheep	sofa	train	tv	mean
MSRA-CFM	-	75.7	26.7	69.5	48.8	65.6	81.0	69.2	73.3	30.0	68.7	51.5	69.1	68.1	71.7	67.5	50.4	66.5	44.4	58.9	53.5	61.8
FCN-8s	-	76.8	34.2	68.9	49.4	60.3	75.3	74.7	77.6	21.4	62.5	46.8	71.8	63.9	76.5	73.9	45.2	72.4	37.4	70.9	55.1	62.2
TTI-Zoomout-16	89.8	81.9	35.1	78.2	57.4	56.5	80.5	74.0	79.8	22.4	69.6	53.7	74.0	76.0	76.6	68.8	44.3	70.2	40.2	68.9	55.3	64.4
DeepLab-CRF	92.1	78.4	33.1	78.2	55.6	65.3	81.3	75.5	78.6	25.3	69.2	52.7	75.2	69.0	79.1	77.6	54.7	78.3	45.1	73.3	56.2	66.4
DeepLab-MSc-CRF	92.6	80.4	36.8	77.4	55.2	66.4	81.5	77.5	78.9	27.1	68.2	52.7	74.3	69.6	79.4	79.0	56.9	78.8	45.2	72.7	59.3	67.1
DeepLab-CRF-7x7	92.8	83.9	36.6	77.5	58.4	68.0	84.6	79.7	83.1	29.5	74.6	59.3	78.9	76.0	82.1	80.6	60.3	81.7	49.2	78.0	60.7	70.3
DeepLab-CRF-LargeFOV	92.6	83.5	36.6	82.5	62.3	66.5	85.4	78.5	83.7	30.4	72.9	60.4	78.5	75.5	82.1	79.7	58.2	82.0	48.8	73.7	63.3	70.3
DeepLab-MSc-CRF-LargeFOV	93.1	84.4	54.5	81.5	63.6	65.9	85.1	79.1	83.4	30.7	74.1	59.8	79.0	76.1	83.2	80.8	59.7	82.2	50.4	73.1	63.7	71.6

Table 3: Labeling IOU (%) on the PASCAL VOC 2012 test set, using the trainval set for training.

Method	mean IOU (%)
MSRA-CFM	61.8
FCN-8s	62.2
TTI-Zoomout-16	64.4
DeepLab-CRF	66.4
DeepLab-MSc-CRF	67.1
DeepLab-CRF-7x7	70.3
DeepLab-CRF-LargeFOV	70.3
DeepLab-MSc-CRF-LargeFOV	71.6

Deep Lab v2

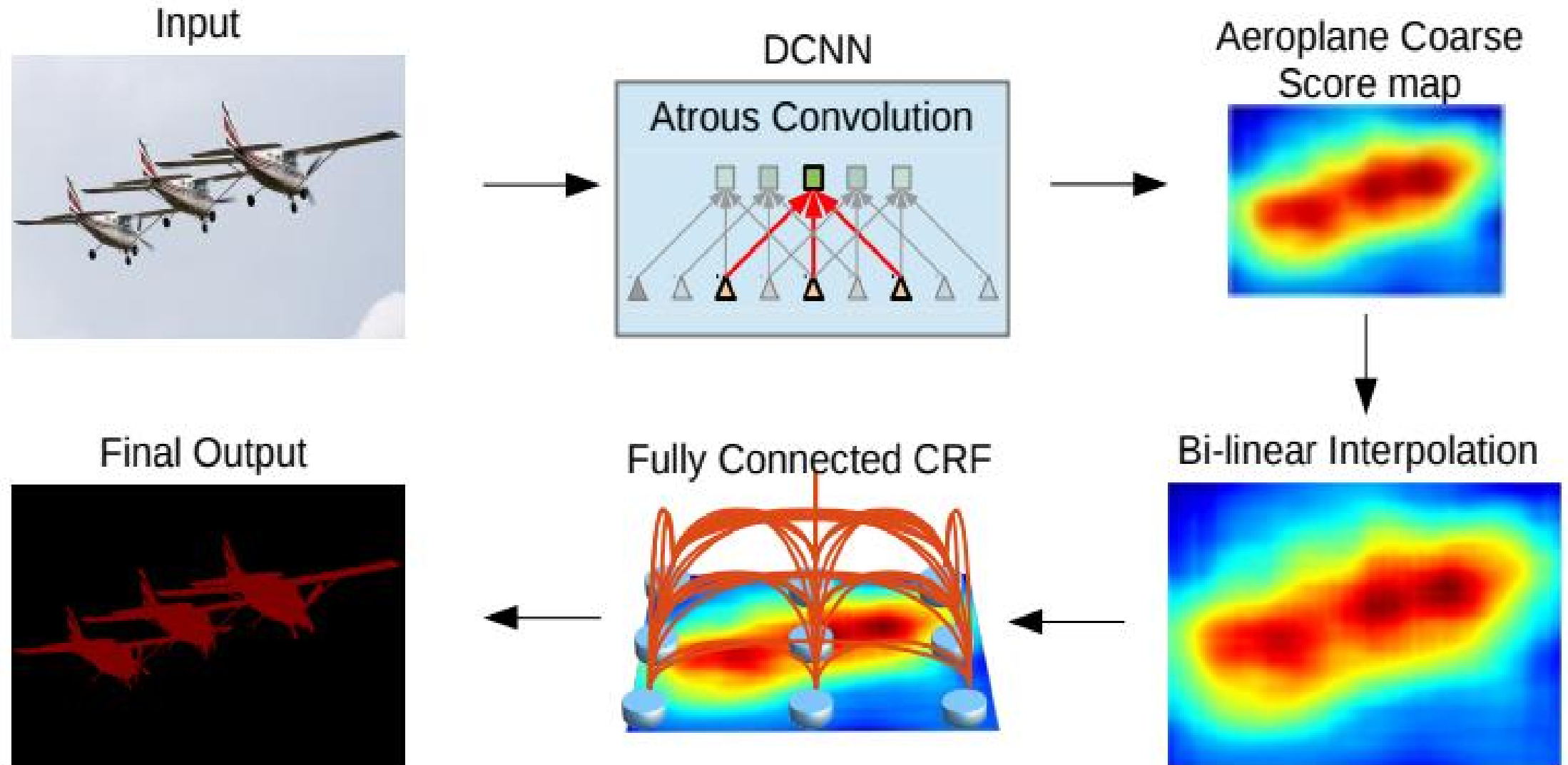
⌚ Deep Lab v2: Semantic Image Segmentation with Deep Convolutional Nets, Atrous Convolution, and Fully Connected CRFs

⌚ <https://arxiv.org/pdf/1606.00915.pdf>

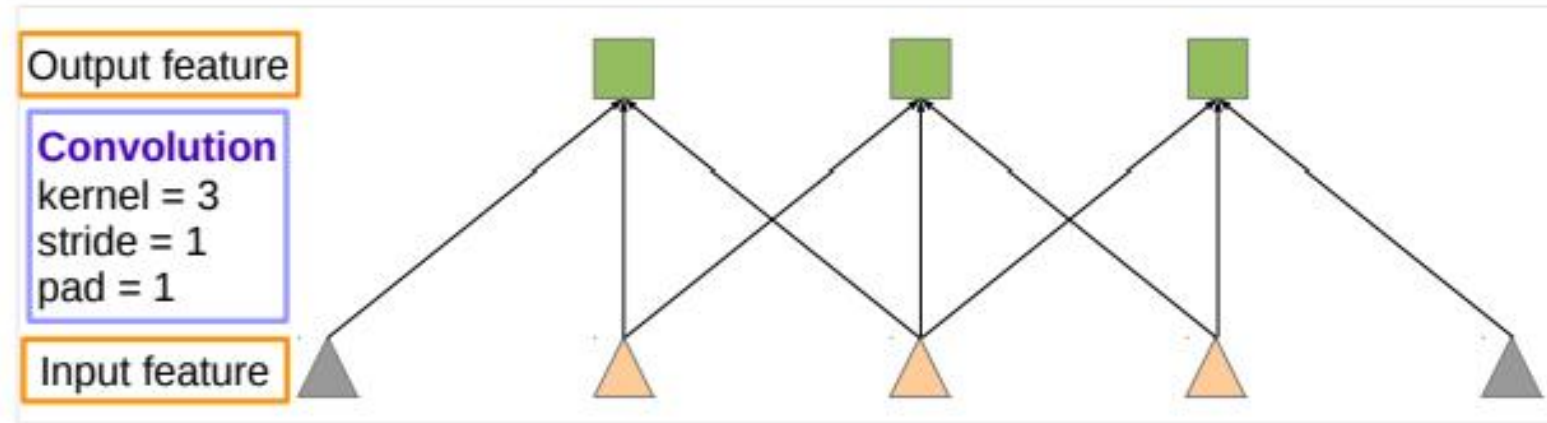
⌚ **Atrous Convolution**

⌚ **Atrous Spatial Pyramid Pooling(ASPP)**

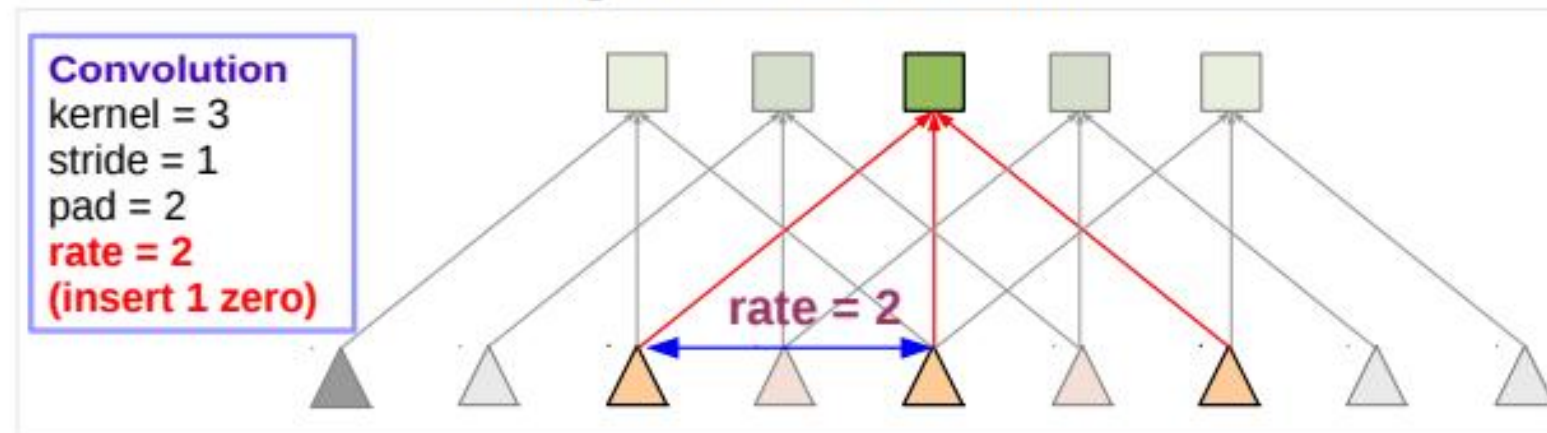
Deep Lab v2



Deep Lab v2

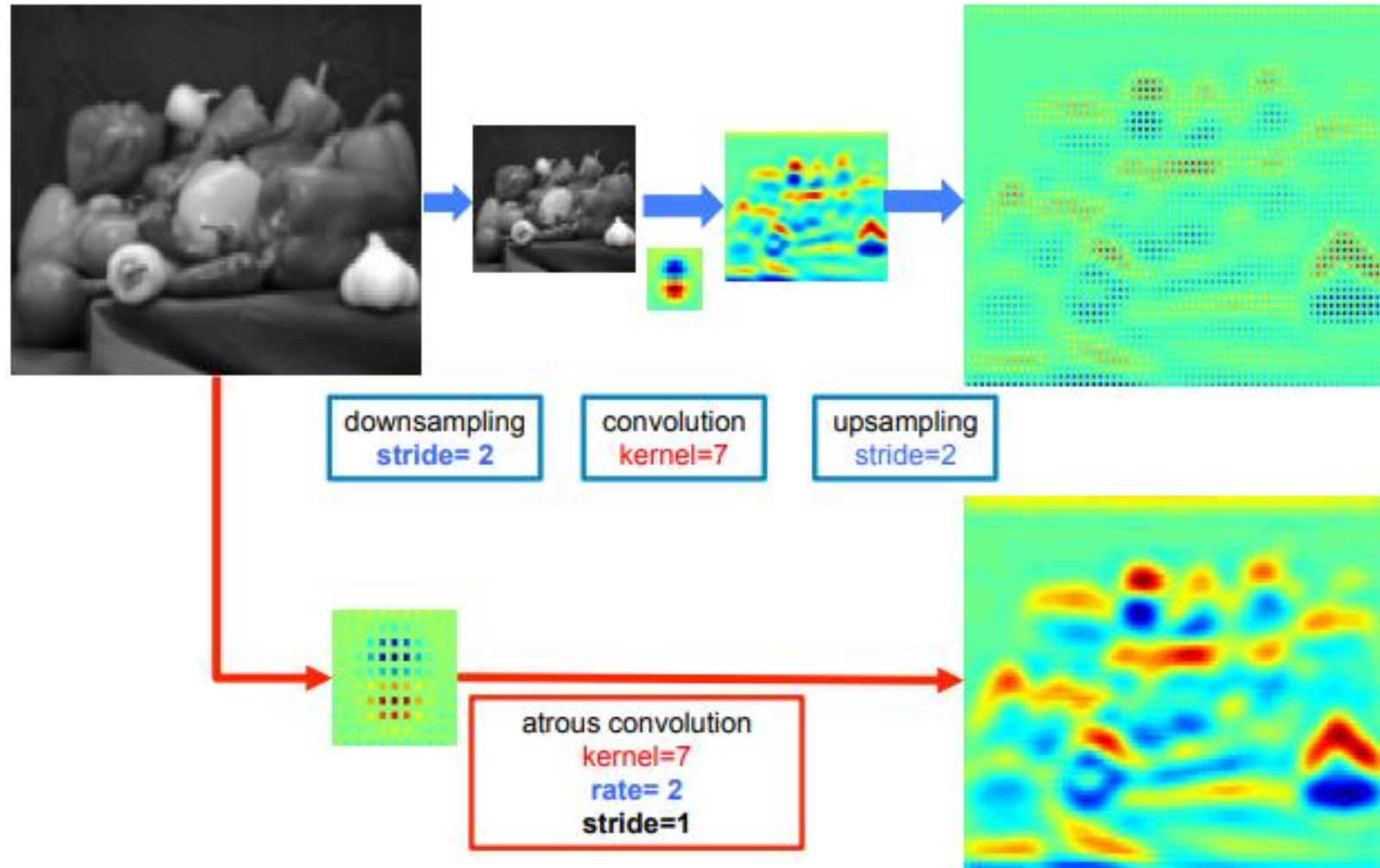


(a) Sparse feature extraction

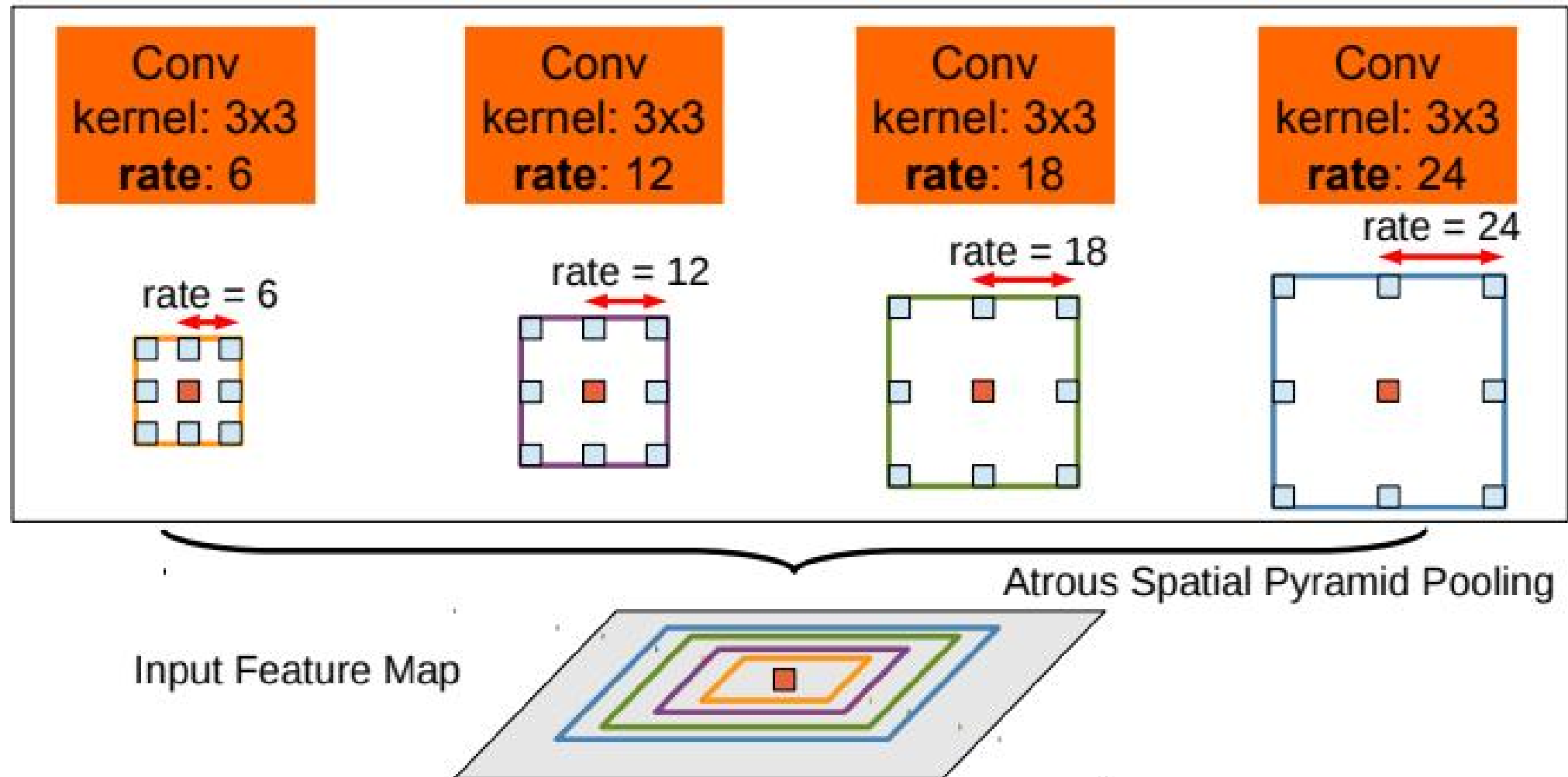


(b) Dense feature extraction

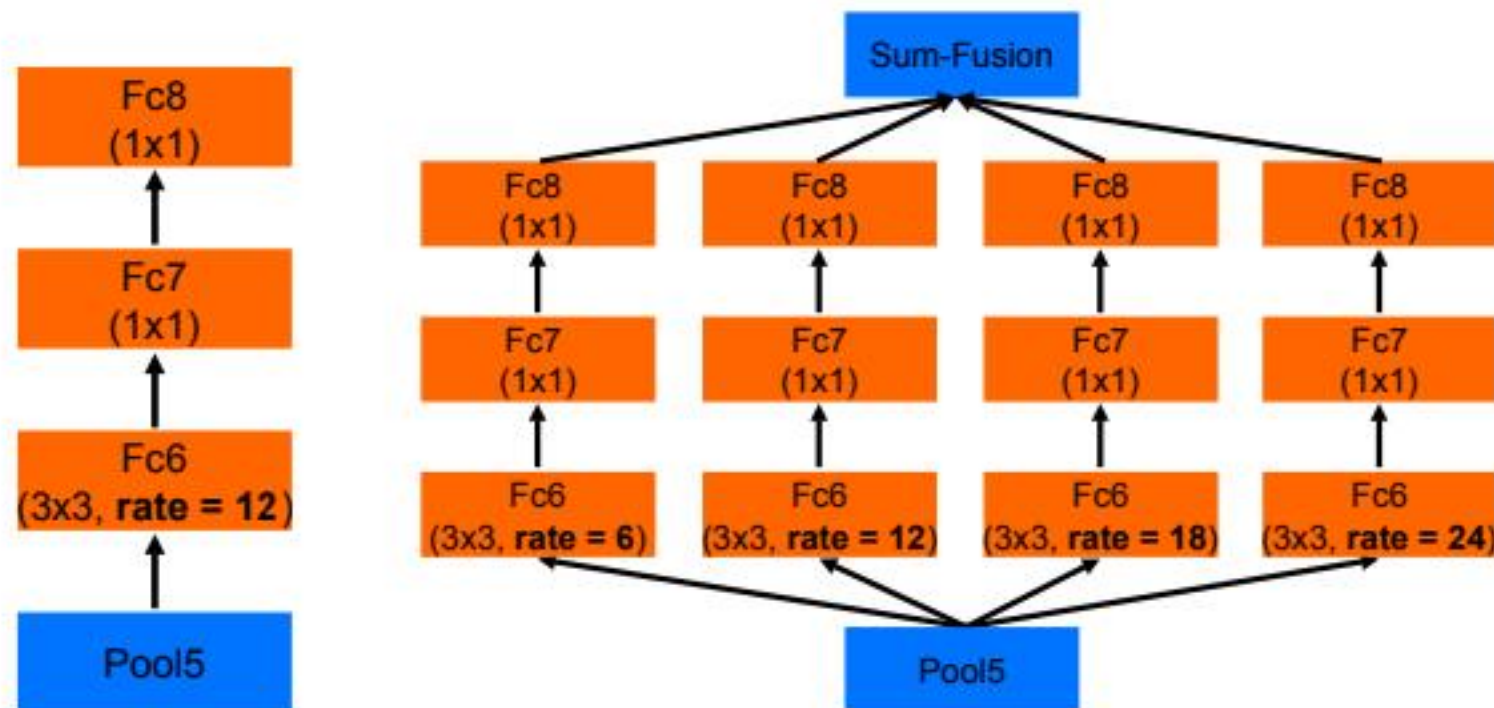
Deep Lab v2



Deep Lab v2



Deep Lab v2



(a) DeepLab-LargeFOV

(b) DeepLab-ASPP

Fig. 7: DeepLab-ASPP employs multiple filters with different rates to capture objects and context at multiple scales.

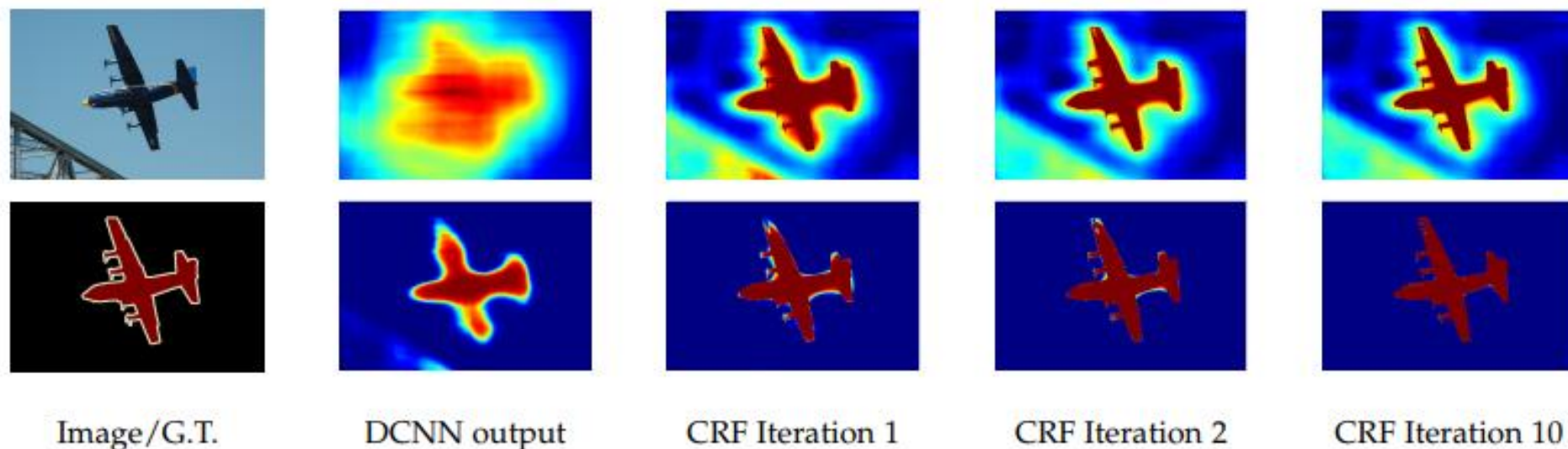


Fig. 5: Score map (input before softmax function) and belief map (output of softmax function) for Aeroplane. We show the score (1st row) and belief (2nd row) maps after each mean field iteration. The output of last DCNN layer is used as input to the mean field inference.

Deep Lab v2

Method	mIOU
DeepLab-CRF-LargeFOV-COCO [58]	72.7
MERL_DEEP_GCRF [88]	73.2
CRF-RNN [59]	74.7
POSTECH_DeconvNet_CRF_VOC [61]	74.8
BoxSup [60]	75.2
Context + CRF-RNN [76]	75.3
QO_4^{mres} [66]	75.5
DeepLab-CRF-Attention [17]	75.7
CentraleSuperBoundaries++ [18]	76.0
DeepLab-CRF-Attention-DT [63]	76.3
H-ReNet + DenseCRF [89]	76.8
LRR_4x_COCO [90]	76.8
DPN [62]	77.5
Adelaide_Context [40]	77.8
Oxford_TVG_HO_CRF [91]	77.9
Context CRF + Guidance CRF [92]	78.1
Adelaide_VeryDeep_FCN_VOC [93]	79.1
DeepLab-CRF (ResNet-101)	79.7

Deep Lab v3

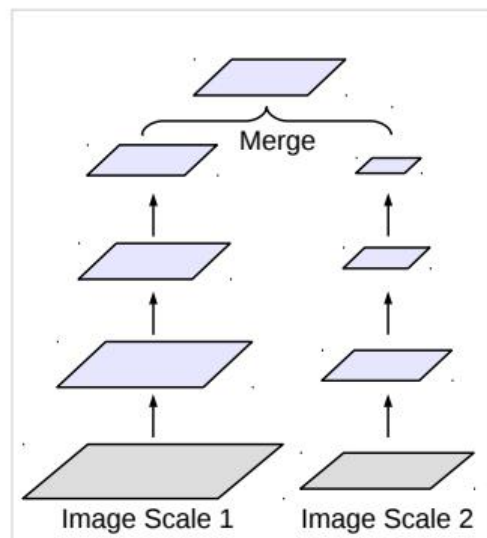
⌚ Rethinking Atrous Convolution for Semantic Image Segmentation

⌚ <https://arxiv.org/pdf/1706.05587.pdf>

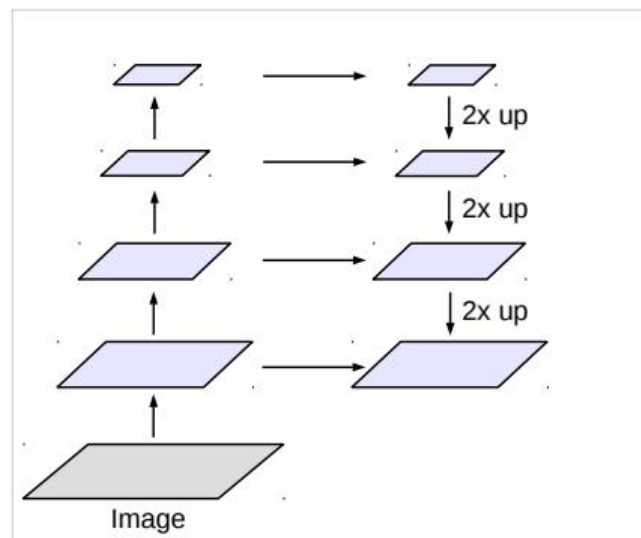
⌚ **Rethinking Atrous Convolution(Atrous Convolution for Dense
Feature Extraction: Multi-Grid & ASPP)**

⌚ **Delete Fully Connected CRF**

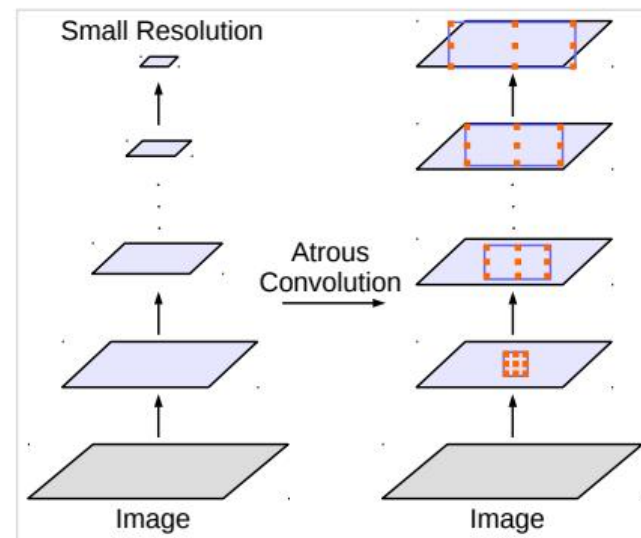
Deep Lab v3



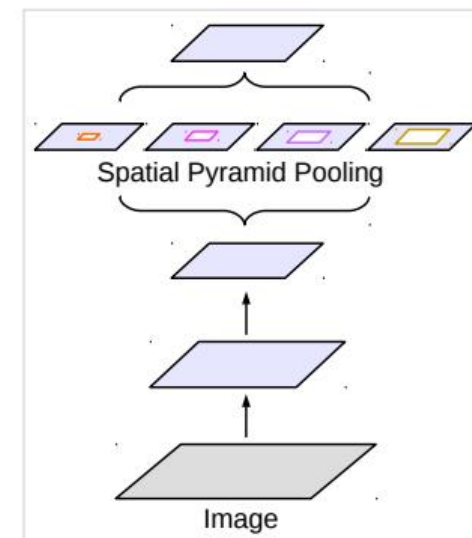
(a) Image Pyramid



(b) Encoder-Decoder



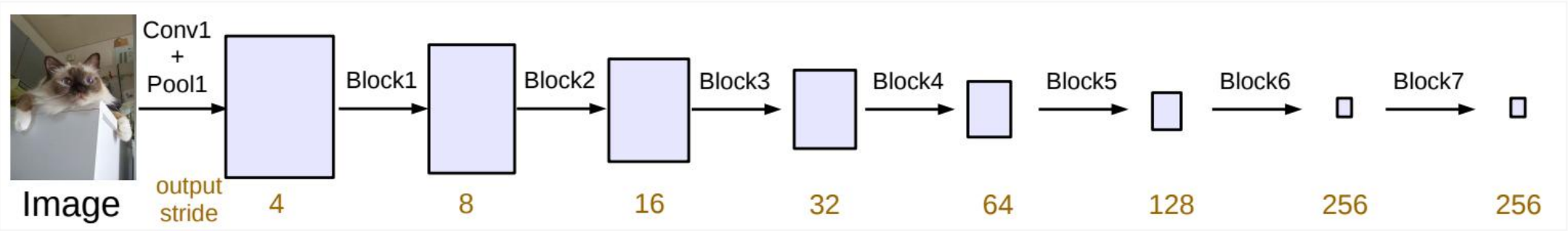
(c) Deeper w. Atrous Convolution



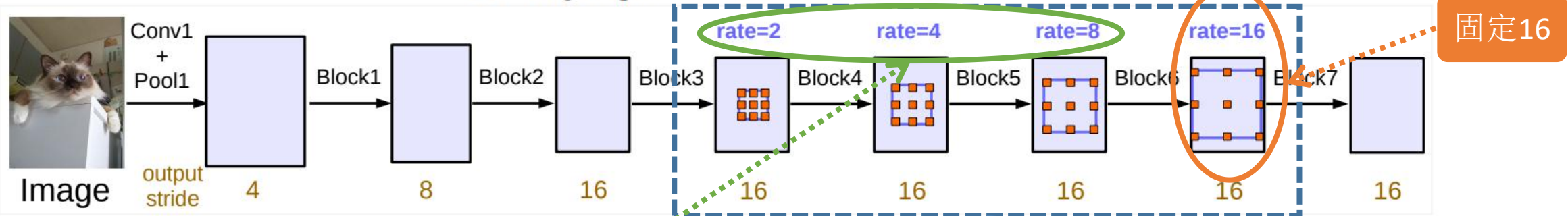
(d) Spatial Pyramid Pooling

Figure 2. Alternative architectures to capture multi-scale context.

Deep Lab v3



(a) Going deeper without atrous convolution.



(b) Going deeper with atrous convolution. Atrous convolution with $rate > 1$ is applied after block3 when $output_stride = 16$.

Figure 3. Cascaded modules without and with atrous convolution.

2*multi-grid;
eg: 2 * (1,2,4)

Atrous Convolution for Dense Feature Extraction: Multi-Grid
使用膨胀卷积提取密集的大视野特征信息
使用膨胀卷积来改善ResNet网络的特征提取功能

Deep Lab v3

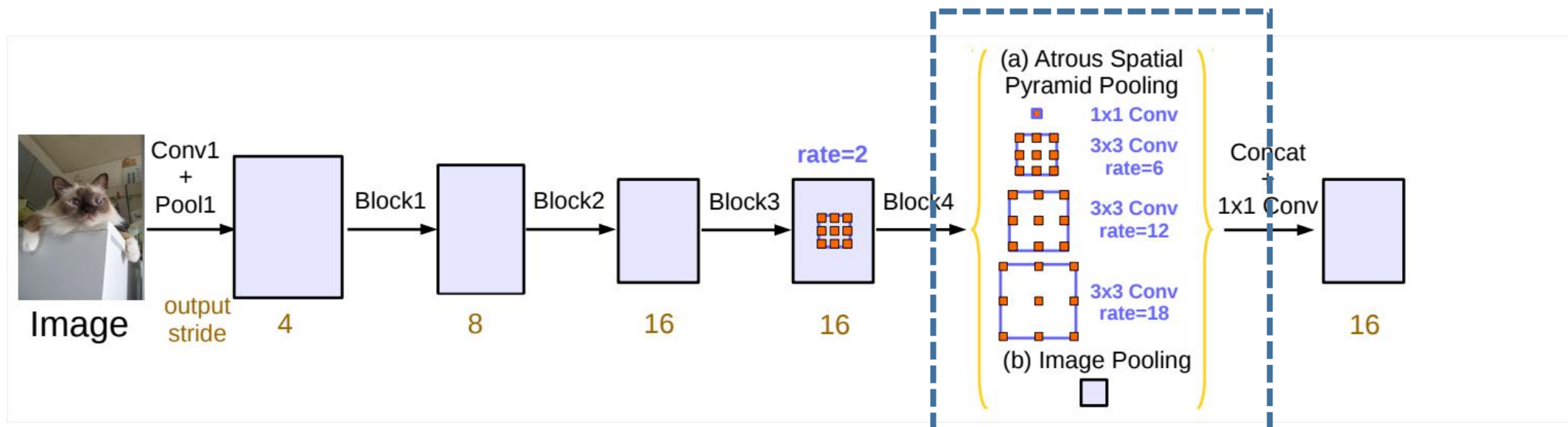


Figure 5. Parallel modules with atrous convolution (ASPP), augmented with image-level features.

Atrous Convolution for Dense Feature Extraction: ASPP
改变ASPP的像素级特征连接方式(sum -> concat)
增加1x1卷积以及多个不同rate的3x3膨胀卷积
膨胀卷积后加入BN
在ASPP中加入普通的Pooling操作

Deep Lab v3

Multi-Grid			ASPP		Image	mIOU
(1, 1, 1)	(1, 2, 1)	(1, 2, 4)	(6, 12, 18)	(6, 12, 18, 24)	Pooling	
✓			✓			75.36
	✓		✓			75.93
		✓	✓			76.58
		✓		✓		76.46
		✓	✓		✓	77.21

Table 5. Atrous Spatial Pyramid Pooling with multi-grid method and image-level features at *output_stride* = 16.

Deep Lab v3

Method	mIOU
Adelaide_VeryDeep_FCN_VOC [85]	79.1
LRR_4x_ResNet-CRF [25]	79.3
DeepLabv2-CRF [11]	79.7
CentraleSupelec Deep G-CRF [8]	80.2
HikSeg_COCO [80]	81.4
SegModel [75]	81.8
Deep Layer Cascade (LC) [52]	82.7
TuSimple [84]	83.1
Large_Kernel_Matters [68]	83.6
Multipath-RefineNet [54]	84.2
ResNet-38_MS_COCO [86]	84.9
PSPNet [95]	85.4
IDW-CNN [83]	86.3
CASIA_IVA_SDN [23]	86.6
DIS [61]	86.8
DeepLabv3	85.7
DeepLabv3-JFT	86.9

Table 7. Performance on PASCAL VOC 2012 *test* set.

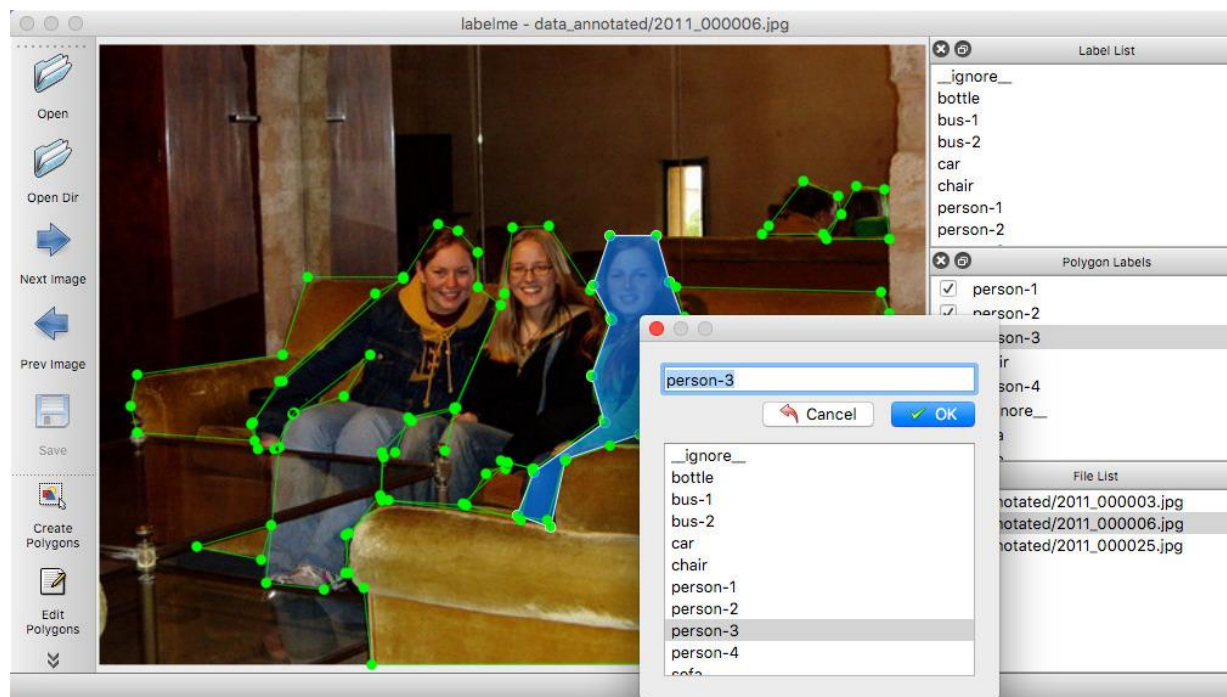
扩展: labelme

⌚ 对于图像分割的数据集构造, 可以考虑使用labelme这个工具进行构造, 详见:

⌚ <https://github.com/wkentaro/labelme>

⌚ 安装方式:

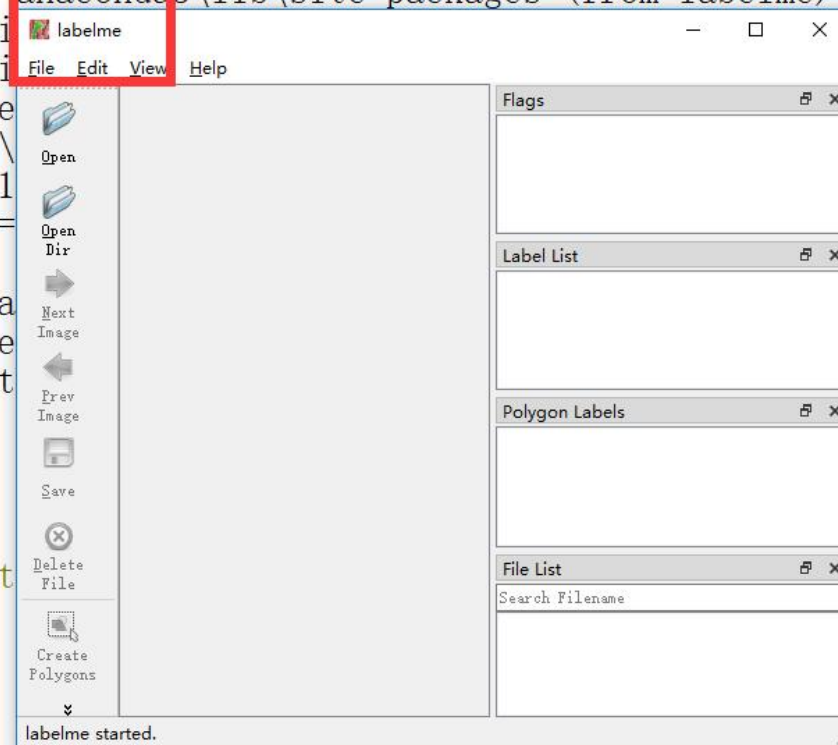
⌚ `pip install labelme`



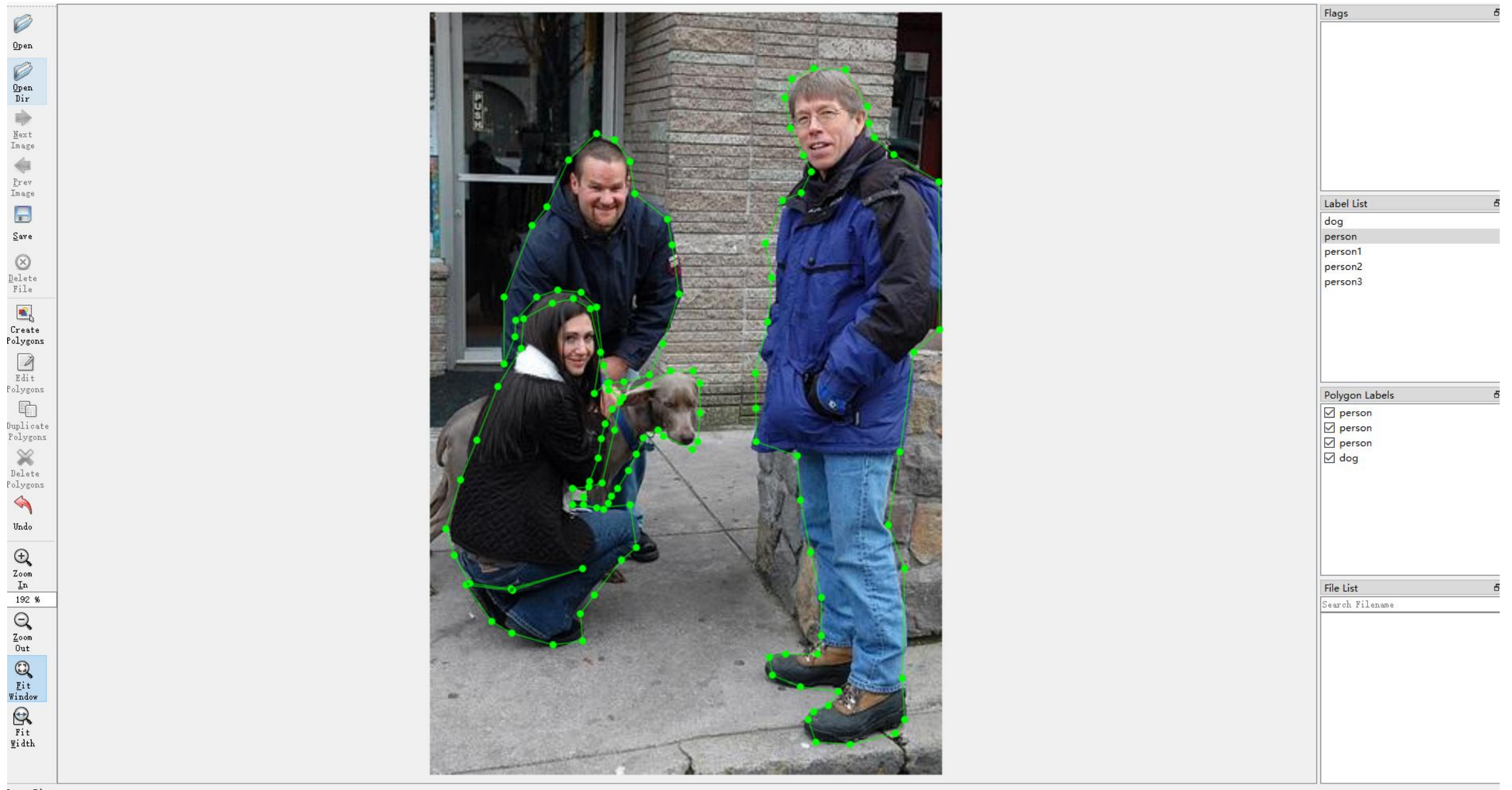
扩展: labelme

```
C:\>pip install labelme
Collecting labelme
  Downloading https://files.pythonhosted.org/packages/8f/71/b8b3b9cb0dbad1658f313ba62f8/labelme-3.16.2.tar.gz (1.5MB)
    100% |#####| 1.5MB 58kB
Requirement already satisfied: matplotlib in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: numpy in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: Pillow>=2.8.0 in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: PyYAML in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: qtpy in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: termcolor in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: six>=1.10 in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: python-dateutil in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: pytz in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: cycller>=0.1 in c:\anaconda3\lib\site-packages (from labelme)
Requirement already satisfied: pyparsing!=2.1.4 in c:\anaconda3\lib\site-packages (from labelme)
Building wheels for collected packages: labelme
  Running setup.py bdist_wheel for labelme
    Stored in directory: C:\Users\user\AppData\Local\pip\Cache\wheels\
Successfully built labelme
Installing collected packages: labelme
Successfully installed labelme-3.16.2
You are using pip version 10.0.1, however
You should consider upgrading via the 'python -m pip install --upgrade pip' command
```

```
>labelme
```



扩展: labelme



扩展: labelme

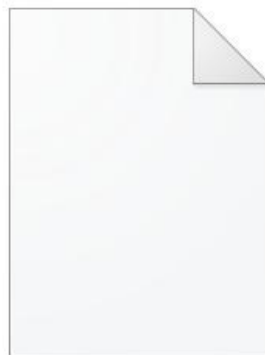
⌚ labelme_json_to_dataset <json文件>

D:\03_数据\VOC2007\VOCtrainval_06-Nov-2007\VOCdevkit\labelme_json_to_dataset 000021.json

```
133m[WARNING] 0m This script is aimed to demonstrate how to convert the JSON file to a single image dataset, and not to handle multiple
JSON files to generate a real-use dataset. (1mjson_to_dataset.py 0m:15)
133m[WARNING] 0m info.yaml is being replaced by label_names.txt (1mjson_to_dataset.py 0m:67)
137m[INFO] 0m Saved to: 000021_json (1mjson_to_dataset.py 0m:72)
```



img.png



info.yaml



label.png



label_names.txt



label_viz.png

扩展: labelme

Open

Open Dir

Next Image

Prev Image

Save

Delete File

Create Polygons

Edit Polygons

Duplicate Polygons

Delete Polygons

Undo

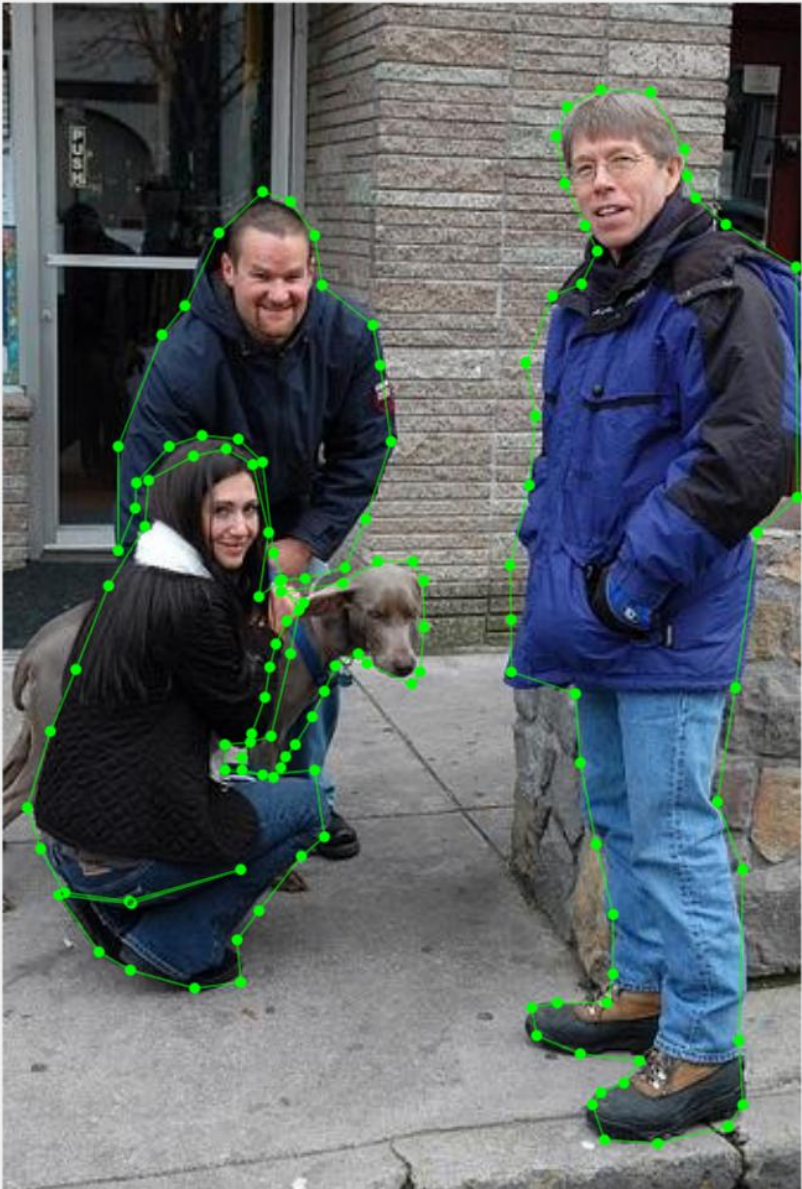
Zoom In

192 %

Zoom Out

Fit Window

Fit Width



Flags

Label List

dog

person

person1

person2

person3

Polygon Labels

☒ person1

☒ person2

☒ person3

☒ dog

File List

Search Filename

扩展: labelme

⌚ labelme_json_to_dataset <json文件>

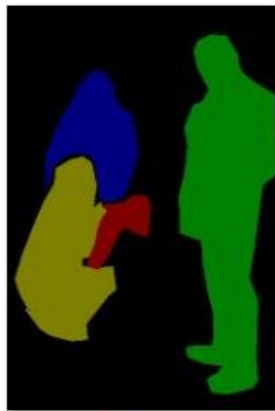
```
D:\03_数据\VOC2007\VOCtrainval_06-Nov-2007\VOCdevkit>labelme_json_to_dataset 000021.json
[33m[WARNING] [0m This script is aimed to demonstrate how to convert theJSON file to a single image dataset, and not to handlemultiple
JSON files to generate a real-use dataset. ([1mjson_to_dataset.py[0m:15)
[33m[WARNING] [0m info.yaml is being replaced by label_names.txt ([1mjson_to_dataset.py[0m:67)
[37m[INFO] [0m Saved to: 000021_json ([1mjson_to_dataset.py[0m:72)
```



img.png



info.yaml



label.png



label_names.txt



label_viz.png

THANKS!